

ORIGINAL ARTICLE

Seropharmacological Study on Osteogenic Effects of Post-Absorption Ingredients of An Osteoprotective Herbal Formula

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ABSTRACT **Objective:** To further investigate the *in vitro* effects of an osteoprotective herbal formula "ELP" (*Herba Epimedii*, *Fructus Ligustri Lucidi* and *Fructus Psoraleae*) using seropharmacological approach. **Methods:** Rats were fed with ELP or its individual component herbs for 2 days. The serum containing the post-absorbed ingredients of the herbal items were collected for cell culture using UMR106 cell, RAW264.7 cell and mesenchymal stem cell (MSC) isolated from the bone marrow of the rats. The effects of the herbal-containing serum on cell toxicity were detected by 3-[4,5-dimethylthiazol-2-yl]-2,5-diphenyltetrazolium bromide assay; bromodeoxyuridine assay was conducted to measure the cell proliferation of UMR106 cell and MSC; cell activity was measured using colorimetric method, and mRNA expression of runt-related transcription factor 2 (Runx2), alkaline phosphatase (ALP) and osteopontin (OPN) of UMR106 and MSC as well as matrix metalloproteinase 9 (MMP-9), tartrate-resistant acid phosphatase (TRAP) and cathepsin K of RAW264.7 were analyzed using real-time reverse-transcription polymerase chain reaction. **Results:** ELP and its component serum exhibited no cytotoxic effects on the cells. The ELP-containing serum increased the proliferation of UMR106 cell and MSC by 25.7% and 14.4%, respectively and the alkaline phosphatase activity of MSC was increased by 42.6%. On the contrary, it inhibited the RAW264.7 cell differentiation by 29.2%. ELP serum upregulated the Runx2 expression of UMR and MSC by 1.18 fold and 1.27 fold, respectively. It also upregulated ALP and OPN expression in MSC by 1.69- and 2.12-fold, respectively. On the other hand, ELP serum down-regulated MMP-9 and cathepsin K expression of RAW264.7 cell by 0.46- and 0.36-fold, respectively. **Conclusions:** The serum of the animals fed with ELP contains active ingredients which are effective in promoting osteogenesis and inhibiting osteoclastogenesis.

KEYWORDS osteoporosis, seropharmacology, osteogenesis, herbal formula

Osteoporosis is one of the commonest bone disorders in the elderly and its associating fractures cause great socioeconomic burdens.^(1,2) Nowadays, pharmaceutical interventions for the treatment of osteoporosis are plentiful.⁽³⁻⁶⁾ However, long-term administration of osteoporotic drugs may cause side effects such as increasing risks of breast and uterine cancer,⁽⁷⁾ venous thromboembolic events,⁽⁸⁾ atypical fracture in long bones^(9,10) and jaw osteonecrosis.^(11,12)

Chinese medicine (CM) is regarded as a safe and effective approach to prevent the development of osteoporosis. Plenty of *in vitro*,⁽¹³⁻¹⁵⁾ *in vivo*⁽¹⁶⁻¹⁸⁾ and clinical⁽¹⁹⁻²¹⁾ studies have been published, and showed that CM was effective for both prevention and treatment of osteoporosis. Nonetheless, in almost all of the *in vitro* studies, researchers put the extracts, developed either from water or ethanol extraction,

into the cell culture system directly. This experimental approach raises doubts on the appropriateness of assuming these *in vitro* results as the real biological effects of the herbs *in vivo*. These extracts have not gone through the process of absorption in the gastrointestinal (GI) tract and subsequent metabolism in the liver. Therefore, it remains controversial that the active ingredients in the herbal extracts affecting the cells *in vitro* could be considered equivalent to

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the post-absorption ingredients of the extracts after digestion. In fact, it should be the active ingredients of the herbal extracts existing in serum treating our diseases. Through Seropharmacology technique, direct observation of the pharmacological effect of the metabolites of herbal medicines after their GI consumption on cells becomes feasible. The *in vitro* results from seropharmacological studies are more appropriate than those from conventional approaches in order to confirm and explain the *in vivo* effectiveness of herbal extracts.⁽²²⁾

A Chinese herbal formula, ELP, is composed of three Chinese medicinal herbs: *Herba Epimedii* (E), *Fructus Ligustri Lucidi* (L) and *Fructus Psoraleae* (P). This herbal formula has been proven effective on the protection against bone loss from osteoporosis through *in vitro*,^(17,23-25) *in vivo*^(16,26,27) and clinical studies.⁽²⁸⁾ ELP is administered orally and is digested in the GI tract, and metabolized in the liver before the active ingredients are transported to the target tissues via blood circulation. It is therefore postulated that the serum of subjects administered with ELP should contain osteoprotective substances.

The objective of this study was to study the osteogenic effects of the post-absorption ingredients of ELP or its individual herbs in the serum.

METHODS

Herbal Materials and Extraction

E, L and P were purchased from a local herbal supplier in Hong Kong, China. Herbarium voucher specimens of the tested herbs were deposited at the museum of the Institute of Chinese Medicine, the Chinese University of Hong Kong, with voucher specimen numbers as: 2004–2547, 2004–2566 and 2004–2568, respectively. The herbs were subjected to extraction using distilled water. Raw materials 250 g of ELP (E:L:P = 5:4:1 w/w); EL (E:L = 1:1 w/w) or single herbal materials (E or L) were soaked in 1-L distilled water for 1 h and then boiled twice for 2 h under reflux. Considering about the hepatotoxicity of P in some individuals when overdosed has been reported,⁽²⁹⁾ it is not recommended for use independently. Therefore, single herb P and its combinations with only E or L had not been studied. The aqueous extract was collected and filtered through a piece of absorbent gauze. The filtrate was then concentrated at 50 °C under reduced pressure, followed by lyophilization in

a freeze dry system into powder form. The powder was finally dissolved in distilled water prior to oral administration.

Animals and Treatment Protocol

A total of 90 male 3-month-old Sprague-Dawley rats (weighing 414.4 ± 27.9 g) were obtained from the Laboratory Animal Services Centre of the Chinese University of Hong Kong. The rats were divided into 5 groups: ELP, EL, E, L and H₂O groups. Each group contained 18 rats. The rats in Group ELP and Group EL were fed with formula ELP and EL, respectively at the dose of 10 g/kg per day (the highest concentration with the viscosity at which the extract could be delivered to the animal fluently) by using gavage tube. The rats in Group E and Group L were fed with E and L, respectively at the dose of 5 g/kg per day (corresponding to their dosage in EL). The concentration of the herbal extracts was adjusted so that each rat received about 4 mL of extract per time. The rats in the control H₂O group were fed with 4 mL distilled water. The feeding was repeated once after 12 h in order to increase the concentration of the post-absorption components of the herbal items in the serum accumulatively. After 2 h of the second feeding, the rats were anesthetized by putting them into a diethyl ether containing chamber until they lost consciousness. Blood was collected from inferior vena cava at the abdominal cavity followed by euthanasia through cervical dislocation. Serum was obtained after the blood samples had been centrifuged at 3,000 r/min for 15 min at 4 °C. Serum from 3 rats in the same group were pooled together into one sample. The pooled serum was then filtered through a 0.2- μ m filter.

Cells and Concentration of Serum

Cell lines including rat osteosarcoma (UMR106) and mouse leukaemic monocyte macrophage (RAW264.7), together with mesenchymal stem cell (MSC) isolated from long bones of rats as described in our prior reports^(17, 25) were involved in this study. UMR106 was cultured in Dulbecco's modified Eagle medium (DMEM; Life Technologies, USA) throughout the study while RAW264.7 was cultured in DMEM for 1 day at the beginning and then the differentiation of osteoclasts from RAW264.7 cells was induced with 50 ng/mL of recombinant mouse soluble receptor activator of nuclear factor kappa B ligand (RANKL; Sigma, USA) in α -methoxyethoxymethyl (α -MEM; Life Technologies) with 10 % fetal bovine serum

(FBS, Life Technologies, USA) for 3 days. MSC was cultured in a growth medium consisting of α -MEM, 10% v/v FBS and 1 % penicillin/streptomycin (Life Technologies).⁽¹⁷⁾ And 10 % v/v rat serum (rat serum to culture medium) were added directly into the cell culture for UMR106 and MSC while 0.15% rat serum was added for RAW264.7.

Cell Viability Assay

The cell viability of the UMR106, undifferentiated MSC or RAW264.7 was determined by using the 3-[4,5-dimethylthiazol-2-yl]-2,5-diphenyltetrazolium bromide (MTT; Sigma, USA) assay after cultured in 96-well plates with serum containing the post-absorption ingredients of the aqueous extracts of the herbs. 1×10^3 cells/well of UMR106 cells, 5×10^3 cells/well of MSC and 500 cells/well of RAW264.7 cells were cultured for 24 h, 48 h and 3 days, respectively. The relative cell viability was determined by measuring the reduction of MTT dye in live cells to blue formazan crystals at optical density at 540 nm and expressed as the percentage of control group without ELP treatment.

Bone Formation Assays

Bromodeoxyuridine (BrdU) assay was conducted to measure the cell proliferation of UMR106 and MSC. 1×10^3 cells/well of UMR106 cells and 5×10^3 cells/well of MSC were seeded in 96-well plates and cultured for 24 and 48 h, respectively. To measure their alkaline phosphatase (ALP) activity, 2.5×10^5 of both cells were seeded in 6-well plates and cultured for 3 and 7 days, respectively. Both the BrdU and ALP activity of the cells were measured using colorimetric methods.^(17,25)

Bone Degradation Assays

RAW264.7 cells (5×10^2) were seeded in each well of a 96-well plate. After treatment with RANKL and different serum for 3 days, the RAW264.7 cell differentiated osteoclasts were fixed and stained for tartrate-resistant acid phosphatase (TRAP) by using an acid phosphatase kit (Sigma) according to the manufacturer's instructions. TRAP-positive multinucleated cells showing more than three nuclei were counted as osteoclasts. Photomicrographs were taken with an inverted microscope at $40 \times$ magnification. The number of osteoclasts per field was calculated by averaging the counting from 8 separated views.

Real-Time Reverse-Transcription Polymerase Chain Reaction

mRNA expression of runt-related transcription factor 2 (Runx2), alkaline phosphatase (ALP) and osteopontin (OPN) of UMR106 and MSC as well as matrix metalloproteinase 9 (MMP9), TRAP and cathepsin K of RAW264.7 were analyzed using real-time reverse-transcription polymerase chain reaction (RT-PCR). 2.5×10^5 UMR106 and MSC, as well as 1.5×10^4 RAW264.7 were cultured in each well of 6-well plates for 24 h, 3 days and 7 days, respectively, before they were harvested for the mRNA expression analysis. The total mRNA of each treated cell culture was isolated by the RNeasy mini kit (Qiagen, GmbH, Germany). The amount of mRNA was determined using the QuantiFast SYBR Green RT-PCR kit (Qiagen, Germany) with a validated primer set from Qiagen specific for the markers stated above on the CFX96 RT-PCR Detection System (Bio-Rad, USA). The threshold cycle (Ct), was determined. Relative expression of the RT-PCR product was calculated using the comparative $2^{-\Delta\Delta C_t}$ method. Endogenous control glyceraldehyde-3-phosphate dehydrogenase mRNA was used for normalization. Fold difference was then determined by normalizing all values to the mean of the relative expression for the control group (H₂O group).

Statistical Analysis

Data were expressed as mean $\pm$ standard error of the mean ($\bar{x} \pm \text{SEM}$). The differences among groups were tested with one-way analysis of variance (ANOVA), followed by the post-hoc Dunnett's test. All statistical analyses were performed using the Statistical Package of Social Science (SPSS) version 15.0 for Windows and carried out at the 5% level of significance ($P < 0.05$).

RESULTS

Herbal-Containing Serum on Cell Viability and Proliferation

Result of the MTT assay showed that there was no significant difference in cell viability when the UMR106 and RAW264.7 were cultured in different rat serum, respectively (Figure). The MSC treated with ELP serum showed 24.0% higher cell viability than that treated with H₂O serum (control) significantly ($P < 0.01$) after cultured for 24 h (Figure 1).

BrdU assay revealed the proliferation rate of UMR106 and MSC elevated significantly when

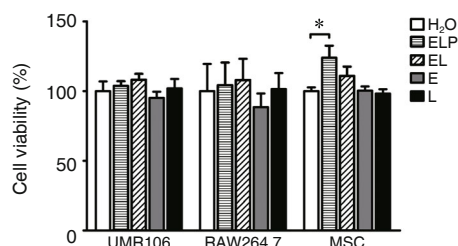


Figure 1. Effects of Different Herbal-Containing Rat Serum on Cell Viability via MTT Assay ($\bar{x} \pm \text{SEM}$)

Notes: Cell viability of the H₂O control was set at 100%.

* $P < 0.01$ vs. H₂O

the cells were incubated with ELP serum (125.7%, $P = 0.011$; 114.6 %, $P < 0.01$, respectively; Figure 2).

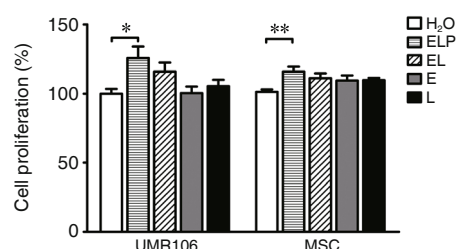


Figure 2. Effects of Different Herbal-Containing Rat Serum on Cell Proliferation through BrdU Assay ($\bar{x} \pm \text{SEM}$)

Notes: Cell proliferation of the H₂O control was set at 100%.

* $P < 0.05$, ** $P < 0.01$, vs. H₂O

Effect of Herbal-Containing Serum on Bone Degradation

The ALP activity of UMR106 which had been cultured for 3 days with the serum from the rats fed with individual herbal items (E: 114.17 U/mg or L: 120.26 U/mg) was higher than the cells cultured with H₂O serum (H₂O: 103.61 U/mg; Figure 3). However, no statistical significance was found. When the MSC was treated with ELP serum, its ALP activity was 42.6% higher than that treated with H₂O serum significantly ($P < 0.05$; Figure 3). The serum of other groups did not affect the ALP activity of MSC.

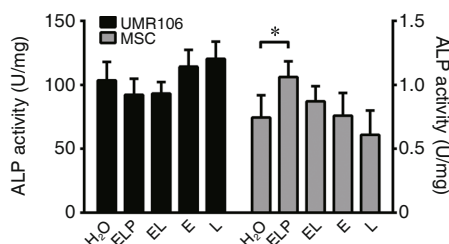


Figure 3. Effects of Different Herbal-Containing Rat Serum on Osteogenic Activity of Cells ($\bar{x} \pm \text{SEM}$)

Notes: ELP Serum did not affect the ALP activity of UMR106 but increased the ALP activity (mean) of MSC.

* $P < 0.05$ vs. H₂O

TRAP staining assay illustrated that the number of osteoclast differentiated from RAW264.7 was reduced when the cells were cultured with serum containing the post-absorption ingredients of the herbal extracts or formulae (Figure 4A). Compared with the H₂O control, ELP, EL and E inhibited the osteoclast differentiation by 29.2 % ($P < 0.05$), 30.9% ($P < 0.05$) and 38.3 % ($P < 0.01$), respectively, after 3 days of culture (Figure 4B).

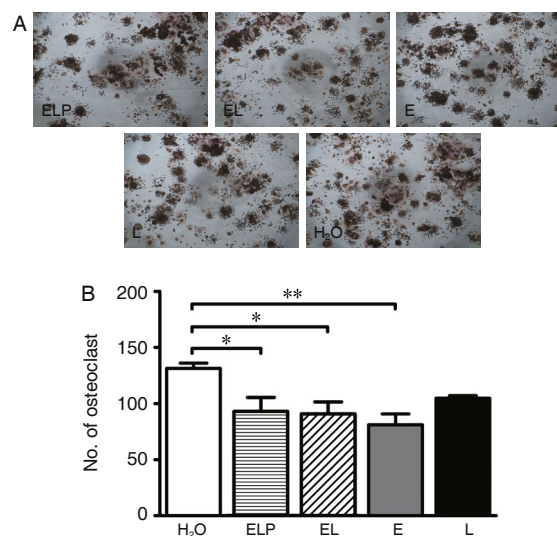


Figure 4. Osteoclast Differentiated from RAW264.7 via TRAP Staining Assay

Notes: (A) Osteoclast differentiated from RAW264.7 ($\times 100$). TRAP-positive multinucleated cells showing more than three nuclei were counted as osteoclasts. (B) The anti-osteoclastogenic effect of the of rats fed with different herbal-containing serum. Mean $\pm$ SEM. * $P < 0.05$, ** $P < 0.01$ vs. H₂O

RT-PCR Results

In the study of UMR106, the Runx2 mRNA expression was upregulated by all the serum containing the post-absorption ingredients of the herbal extracts when compared with the H₂O control (Figure 5). The fold of increase in group ELP, EL, E and L was 1.18 ($P = 0.048$), 1.57 ($P < 0.01$), 1.12 ($P = 0.025$), and 1.23 ($P = 0.015$), respectively. However, the ALP expression had not been affected (Figure 5). For the OPN mRNA, EL serum showed a significant up-regulatory effect on its expression (1.20-fold, $P < 0.01$; Figure 5).

In the MSC study, the expressions of the Runx2, ALP and OPN mRNA were all up-regulated by the ELP serum when compared with the H₂O control by 1.27-fold ($P < 0.01$), 1.69-fold ($P < 0.01$) and 2.12-fold ($P < 0.05$), respectively (Figure 6). Runx2 was also upregulated by the E serum significantly by 1.29-fold ($P < 0.01$; Figure 6), whereas L serum also upregulated the ALP mRNA

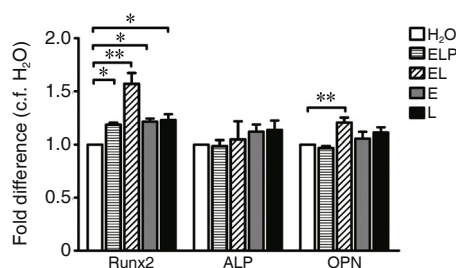


Figure 5. Different Herbal-Containing Rat Serum on mRNA Expression of UMR106 ($\bar{x} \pm \text{SEM}$)

Notes: All of the rat serum upregulated the Runx2 expression and EL serum upregulated the OPN expression. Results are the fold difference from control (H₂O). * $P < 0.05$, ** $P < 0.01$ vs. H₂O

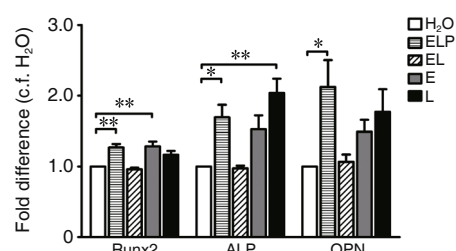


Figure 6. Effects of Different Herbal-Containing Rat Serum on mRNA Expression of MSC ($\bar{x} \pm \text{SEM}$)

Notes: ELP Serum upregulated all Runx2; ALP and OPN expression of MSC. * $P < 0.05$, ** $P < 0.01$ vs. H₂O

expression by 2.04-fold ($P < 0.01$; Figure 6).

MMP-9 and cathepsin K mRNA expression of RAW264.7 were downregulated significantly by the ELP serum 0.46 ($P < 0.05$) and 0.36 ($P < 0.01$) fold, respectively. There was no significant difference in TRAP expression of RAW264.7 by the post-absorption ingredients of the herbal extracts when compared with the H₂O control (Figure 7).

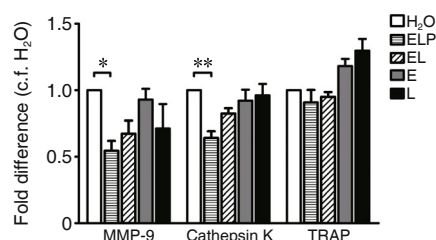


Figure 7. Effects of Different Herbal-Containing Rat Serum on mRNA Expression of Osteoclast ($\bar{x} \pm \text{SEM}$)

Notes: ELP Serum downregulated MMP-9 and Cathepsin K expression. The TRAP expression had not been affected. * $P < 0.05$, ** $P < 0.01$ vs. H₂O

DISCUSSION

The results from seropharmacological study are more appropriate to explain the pharmacological mechanisms of complex chemical mixtures like herbal

extracts with regard to their enzymatic, molecular and genetic effects than those observed from direct conventional cell cultures. It also reduces the interference of the physical factors (such as variation of purity, pH value) of crude extract on cells in *in vitro* studies. More importantly, the main components or the active ingredients that are the most effective to treat the target disease can be identified easier through analyzing the post-absorption ingredients of the extracts in serum than through the crude extracts. This provides evidences to facilitate the further development of CM. Furthermore, it can also be used to study the absorption and metabolism pathways of the CM easily. Summarizing the above advantages, seropharmacology can act as a connection between the classical wisdom of CM and the modern concepts, technologies and theories of life sciences.

From the current study, it was found that the rat serum containing the post-absorption ingredients of the herbal extracts did not exhibit cytotoxic effect on the cells through the MTT assays. Both UMR106 and MSC could sustain 10% of rat serum during the cell culture, although RAW264.7 could only sustain 0.15%, even in the serum of the rat fed with distilled water. It reflected that the culture of RAW264.7 is sensitive to rat serum. The results also demonstrated a critical protocol for the culture of RAW264.7 for seropharmacological experiments in the future.

ELP was found to be more potent than EL, E and L in osteogenesis. From the point of view of osteoblastogenesis, significant increase in cell proliferation on UMR106 and MSC was found in the cell treated with ELP serum only via BrdU assay. Enhancement of ALP activity was also observed when MSC had been cultured in ELP serum, but not in other serum. Considering from osteoclastogenesis, although all the serum containing herbal formulae or individual herbal extracts, except the L serum, led to a significant inhibitory effect on osteoclast formation, only ELP serum downregulated the MMP9 and cathepsin K expression in RAW264.7 significantly. Therefore, ELP post-absorption ingredients are believed to reduce the recruitment of osteoclasts for bone resorption and reduce their bone resorbing activity as well.

The osteogenic effect of the ELP serum might be contributed by the osteogenic effects of the active ingredients of ELP. They include icariin, its metabolites

icaraside I and icaraside II; salidroside, oleanolic acid and ursolic acid, psoralen and isopsoralen. In fact, one of our previous studies showed that the aqueous extract of ELP contains icariin, salidroside, psoralen and isopsoralen.⁽³⁰⁾ Another study revealed that icaraside II, salidroside, psoralen and isopsoralen were found in the serum of the rats fed with ELP (unpublished data). Icariin and its metabolites have been well accepted as the promoting substances for bone formation^(15,31-33) and bone resorption.⁽³⁴⁻³⁶⁾ Salidroside has also been found as the active ingredient of *Fructus Ligustri Lucidi*, which exhibits not only osteogenic effect, but also anti-osteoclastogenic effect *in vitro* and *in vivo*.^(37,38) The osteogenic effect of psoralen has also been demonstrated through *in vitro* and *in vivo* studies.^(39,40)

In an *in vivo* study investigating the efficacy of ELP on prevention of osteoporosis development due to weightlessness using tail-suspended rats, ELP was found reducing the level of bone turnover markers in serum of the rats in a dose-dependent manner. After ELP administration for 28 days, the reductions in bone formation markers—bone-specific alkaline phosphatase and osteocalcin, as well as bone resorption marker—tartrate-resistant acid phosphatase 5b, in the serum, increased as the dosage of ELP increased (unpublished data). These results indicated that ELP reduced the bone turnover rate of the tail-suspended rats. However, the dose-dependent increase in bone mineral density of the rats illustrated that the ELP caused a net bone gain to the rats during the suspension period.⁽¹⁷⁾ These observations were coherent with the results of another *in vivo* study. *Herba Epimedii* extract reduced both bone formation and bone resorption markers in the serum of the ovariectomized rats but prevent ovariectomy-induced bone loss through histomorphometric analysis.⁽²³⁾

ELP was also more comprehensive on osteogenesis of MSC derived osteoblast rather than of UMR106. When focus was put on analyzing the effect of ELP post-absorption ingredients on osteoblastogenesis, although ELP serum showed its osteogenic properties in term of proliferation on both UMR106 and MSC as stated above, its promoting effect on osteogenic activity was limited to MSC only. The elevated ALP activity by ELP serum was observed in MSC only, but not in UMR106. These results were coherent with the findings from the analysis on gene expression of these cells. Runx2, the master gene for initiation of osteoblast differentiation, was significantly upregulated in UMR106 and MSC by

the ELP serum. However, other genes relating to the bone forming activity of osteoblasts, ALP and OPN, had not been promoted in UMR106. On the contrary, they had been upregulated in MSC significantly.

Notably, the increase in ALP activity of MSC and reduction in osteoclast number differentiated from RAW264.7 by the post-absorption ingredients of ELP were comparable to one of our previous reports. In which, the aqueous extract of ELP enhanced the ALP activity of MSC and reduced the osteoclast formation in a dose-dependent manner.⁽¹⁷⁾ These observations revealed that the herbal formula ELP, no matter its crude extract or its post-absorption ingredients in serum, can promote the osteogenic activity of osteoblast derived from MSC and can also inhibit osteoclastogenesis.

Besides, L containing serum caused the highest ALP activity of UMR106 among the other groups at 3 days of cell culture. It also reduced the number of osteoclast differentiated from RAW264.7, although all of these were not statistically significant compared with the control serum. Remarkably, it up-regulated the ALP gene expression of MSC significantly. Summarization of these observations leads to a consideration that the post-absorption ingredients in L serum should be effective on osteogenesis but they were not potent enough when studying on cells *in vitro*. The concentration of active ingredients of a herb ought to be reduced after undergoing digestion and metabolization when compared with its crude extract. In fact, the inhibitory effect of aqueous extract of L on osteoclastogenesis had been proven in a dose-dependent manner in one of our previous studies.⁽²⁵⁾ Aqueous extract of L could increase the OPG/RANKL ratio in MSC significantly when its concentration was at least 50 μ g/mL. The promoting efficacy of aqueous extract of L on proliferation of MSC and ALP activity of MSC is also dose-dependent.^(25,41,42) Having not measured the concentration of the marker compounds in the serum was one of the limitations in the current study.

From this study, we could also find that there is no synergistic osteogenic effect when E combines with L only. However, ELP was the most effective herbal item among the four on stimulating osteoblastogenesis and on inhibiting osteoclastogenesis. Therefore, it is interesting to hypothesize that whether or not the presence of P, although it occupies only 10% weight/weight in the formula, plays an important role for ELP

to treat osteoporosis. Further studies on investigating the mechanisms of P on etiology of osteoporosis *in vitro* are worth to be conducted although a side effect of P associating with hepatotoxicity in some individuals when overdosed has been reported.⁽²⁹⁾

The seropharmacological approach in studying CM has not been widely adopted till now. The study of efficacy of CM on osteoporosis through this approach is even less. The results of this study reveal that seropharmacology can exhibit the direct effects of the post-absorption ingredients of herbal items on cells. It offers a convenient platform of research in which the unnecessary compounds are eliminated, while the bioactive compounds persist for biological studies. The results thus obtained are more relevant than the conventional *in vitro* studies in an attempt to explain the *in vivo* efficacy of herbal medicine.

Conflict of Interests

The authors have declared that there is no conflict of interest.

Author Contributions

Ko CH, Hung LK and Leung PC designed the experiments. Lau CBS contributed to the experimental design and manuscript editing. Siu WS, Wong HL, Gao S and Shum WT conducted the experiments. Siu WS and Ko CH performed the analysis. Siu WS drafted the manuscript. All authors read and approved the final version of the manuscript.

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